Succession Over Accumulation: Feedforward as the Structural Mechanism of Continuity

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Abstract

This paper proposes that continuity is not the minimisation of reconstruction cost but the elimination of the reconstruction event itself — and that feedforward, not feedback, is the only control mechanism structurally capable of achieving this, because feedback by definition operates after the disturbance it corrects for. Drawing on Wiener's foundational distinction between feedforward and feedback control, and on the Conant-Ashby Good Regulator Theorem, the paper argues that the unit of successful continuity is succession — the handoff of a working model of prior reasoning from one reasoning vehicle to the next — rather than accumulation of stored content. The paper distinguishes succession from accumulation precisely, explains why accumulation without feedforward structurally guarantees reconstruction, and proposes epistemic maturity as a necessary property of any artifact intended to carry reasoning forward. The argument is presented as a navigational hypothesis, grounded in established cybernetic theory and illustrated — not proven — by an exploratory cross-instance AI experiment.

Keywords: feedforward, feedback, good regulator theorem, requisite variety, succession, navigational coherence

I. The Quiet Assumption Behind Most Continuity Infrastructure

Most attempts to solve institutional knowledge loss share an unexamined assumption: that the problem is retrieval. If the right information can be stored, indexed, and made searchable, continuity follows.

This assumption produces an enormous and growing industry — document management, knowledge bases, version control, organisational wikis, increasingly sophisticated retrieval-augmented AI systems. All of it optimises for the same underlying goal: when a future person or system needs to know something, make it findable, and make finding it cheap.

This is feedback logic. It is reactive by construction. It waits for the need to arise, then reduces the cost of locating an answer. Even at its most sophisticated, it is a correction mechanism applied after a gap has already opened between what was once understood and what is currently accessible.

The previous papers in this series named that gap. Interpretive Entropy describes its accumulation (Mayoh, 2026a). The Continuity Threshold describes the point past which it becomes self-reinforcing (Mayoh, 2026b). Invisible Opportunity Cost describes what is lost while nobody is measuring (Mayoh, 2026c, forthcoming).

Persistent Semantic Scaffolds, as a proposed architecture, describes how to contain the damage once a system has grown complex enough that the gap is structurally inevitable (Mayoh, 2026d, forthcoming).

All of this is, in cybernetic terms, feedback architecture — increasingly well-designed feedback architecture, but feedback nonetheless. It detects drift. It contains mutation. It lowers the cost of reconstruction.

None of it asks a more basic question: can the reconstruction event be prevented from occurring at all?

II. Wiener's Distinction

Norbert Wiener founded cybernetics in 1948 around a single observation that has only grown more relevant since: purposeful systems regulate themselves through information about their own behaviour, and that information can arrive in two structurally different ways.

Feedback systems monitor output after it has occurred and adjust based on the deviation from a desired state. A thermostat that turns on heating once a room has already cooled is operating on feedback. The correction always trails the disturbance.

Feedforward systems act on a disturbance before it has affected the system, using a prediction of what is about to happen rather than a measurement of what already has. If you perceive a sudden movement near your face, you close your eyelids before any projectile can hit it — preventing the deviation rather than correcting it afterward.

The distinction is not a matter of degree. It is structural. Feedback can only ever be as fast as the disturbance it is correcting for. Feedforward can, in principle, prevent the disturbance from mattering at all — at the cost of requiring an accurate anticipatory model, which feedback does not need.

Subsequent work in cybernetics and control theory has consistently found that the most robust systems combine both: a feedforward controller that monitors potential disturbances at runtime and uses predefined anticipatory controls to preempt them directly, paired with a feedback controller that monitors measured output and adjusts according to the difference from a reference value. Neither mechanism alone is sufficient. But they are not interchangeable, and a system designed only for feedback inherits feedback's fundamental limitation: it cannot act before the deviation it is meant to address.

This is the limitation every purely reconstruction-cost-reducing architecture inherits. No matter how cheap reconstruction becomes, it remains reconstruction. The event it was meant to prevent has already occurred.

III. The Good Regulator Theorem

The formal justification for why feedforward requires more than a record of conclusions comes from a result in cybernetics that is, to date, underused outside control theory and management science.

In 1970, Roger Conant and W. Ross Ashby proved what has become known as the Good Regulator Theorem: every good regulator of a system must be a model of that system. To have sufficient variety to regulate a system, the regulator must, in some functional sense, contain a representation of what it's regulating — a model that allows it to anticipate the system's behaviour and generate appropriate responses.

The implication, stated as plainly as the theorem allows: you cannot control what you cannot predict. You cannot predict what you have not modeled.

This is not a metaphorical claim. It is a mathematical result about the structure of control systems, with applications ranging from biological regulation to organisational management. An animal's brain must recognise and remember the state of things it monitors and directs; a business must recognise the state of things it monitors and directs — in each case, the sensation or memory must model or represent reality well enough for the system to act effectively and survive.

Applied to reasoning continuity, the theorem has a precise and somewhat unforgiving consequence: a record that preserves only conclusions cannot function as a regulator of future reasoning, because a conclusion is not a model. It is the output of a model, stripped of the model itself. A future reasoning vehicle — human or AI — inheriting only a conclusion has no way to anticipate how that conclusion should be extended, revised, or applied to a situation the original reasoning did not explicitly address. It can only ever reconstruct the model from the conclusion backward, which is precisely the expensive, lossy, interpretive-entropy-generating process this entire research series has been describing.

A record that preserves intent, assumptions, reasoning structure, and constraints — rather than conclusions alone — is doing something categorically different. It is transmitting a partial model. It gives the next reasoning vehicle enough structure to predict how the original reasoning would extend, not merely what it once concluded.

This is the theoretical justification this series has, until now, been missing for why the Pay-It-Forward Record was structured the way it was structured. The PIFR's specification — intent, assumptions, reasoning steps, constraints, conclusions, criteria for future continuation — is not an arbitrary list of useful fields. It is, in Conant and Ashby's terms, an attempt to transmit a sufficient model, not merely a sufficient answer.

IV. Requisite Variety and the Thinness Problem

Ashby's earlier and more general result, the Law of Requisite Variety, adds a further constraint. The wider the variety of states a regulator must control, the wider the variety of states it must recognise, and the wider the variety of actions it must be able to perform.

This explains a failure mode that anyone who has worked with thin documentation will recognise immediately. A record that captures only a single dimension of prior reasoning — only the decision, or only the data, or only the final design — has insufficient variety to regulate the wide range of future situations a reasoning vehicle will actually encounter. It can answer the narrow question it was built to answer. It cannot anticipate the adjacent questions that inevitably arise once the original context has been forgotten.

This is why accumulation, on its own, does not solve continuity even when it produces enormous volumes of stored content. More documents do not mean more requisite variety. A thousand records of conclusions, with no model behind any of them, still leave the regulator — the next person or AI inheriting the work — with insufficient variety to anticipate anything the records did not explicitly anticipate for it.

Variety has to be designed into the record, not assumed to emerge from its volume.

V. Succession, Not Accumulation

This gives the paper its central distinction.

Accumulation is the process of adding more stored content over time: more documents, more decisions logged, more conclusions recorded. Accumulation answers the question *how much do we have*.

Succession is the process by which one reasoning vehicle hands a working model of its reasoning to the next, such that the next vehicle can continue without reconstructing the original context. Succession answers a different question entirely: *can whoever inherits this continue from where I left off, oriented as I was oriented*.

These are not different amounts of the same thing. They are structurally different processes, and a system can accumulate enormous amounts of content while achieving almost no succession at all — which is precisely the condition this series has called Interpretive Entropy at scale.

The organisational succession planning literature, working entirely independently of this series, has already converged on the same distinction in human institutions. Succession planning facilitates the transfer of tacit knowledge — the wisdom gained through experience — from one generation of workers to the next, shortening the learning curve for successors and reducing the need for resource-intensive formal training. Nonaka's widely used SECI framework treats the fluid transfer of tacit and explicit knowledge among individual, group, and organisational levels as the actual mechanism of organisational continuity — not the size of the organisation's documentation.

A finding from the enterprise knowledge management literature states the failure mode almost exactly as this paper would state it: many organisations focus solely on documenting step-by-step processes, overlooking the tacit knowledge that explains the "why" behind key decisions — the critical context future employees actually need (Nonaka & Takeuchi, 1995).

This is the case for feedforward, arrived at independently by practitioners solving a narrower, more immediate version of the same structural problem: documentation without a model of the reasoning behind it produces accumulation, not succession.

VI. Why Mindless Accumulation Always Produces More Reconstruction

The mechanism can now be stated precisely.

A system optimised purely for accumulation has no mechanism for asking, at the moment content is added, whether that content is legible to whoever inherits it next. Each addition is judged successful if it is added — stored, indexed, retrievable. Nothing in the accumulation logic asks whether the addition carries a model sufficient for future regulation, in Conant and Ashby's sense.

The result is structurally guaranteed, not accidental. As content accumulates without an accompanying model, semantic surface area grows — more terms, more cross-references, more historical decisions sitting in the record — while the requisite variety needed to navigate that surface area does not grow proportionally, because nothing in pure accumulation logic was building it. The gap between what has been stored and what can actually be regulated by a future reasoning vehicle widens with every addition.

This is reconstruction cost increasing not despite accumulation but because of it. More stored content, with no feedforward structure, does not make a system more navigable. It makes the eventual reconstruction event larger.

The phrase that captures this most simply: *we have done a lot for more. The way forward is more sensible.* Sensible accumulation is accumulation that asks, at the point of creation, who inherits this and how will they continue it — the second question pure accumulation never asks. A Pay-It-Forward Record is what "more sensible" looks like in concrete form: not a smaller amount of content, but content deliberately structured to carry a model forward rather than merely a conclusion.

VII. Epistemic Maturity as a Necessary Property

One further property is required for a feedforward record to function safely, which the original PIFR specification implies but does not yet make explicit.

A model transmitted forward is only useful to a future regulator if the regulator also knows how confidently to hold it. The same conclusion, carried forward as an established finding versus carried forward as an untested hypothesis, should be treated completely differently by whoever inherits it — built upon directly in the first case, tested and scrutinised in the second.

Without an explicit marker of epistemic maturity, these two cases are indistinguishable in the record. A future reasoning vehicle — human or AI — has no way to know whether it is inheriting solid ground or a working theory dressed in the same format as solid ground. This is a specific and underappreciated form of interpretive entropy: confident-sounding structure smuggling an untested claim into the status of settled fact, simply because the artifact carrying it does not distinguish between the two.

The proposal here is narrow and practical: every feedforward artifact should carry an explicit maturity marker — something as simple as hypothesis, working theory, tested, or established — attached at the point of conclusion. This does not solve epistemic uncertainty. It makes the uncertainty itself part of what is transmitted, which is precisely what Ashby's variety requirement demands: the regulator needs to know not only what was concluded, but how much weight that conclusion can bear.

The marker alone is not sufficient, and this is worth stating precisely rather than leaving implicit. A marker is only checkable — as opposed to merely asserted — if the artifact also records what would need to be true for that level of confidence to be warranted. Persistent Semantic Scaffolds and the Pay-It-Forward Record (Mayoh, 2026d, forthcoming) specifies **criteria for success** as a PIFR field, recommended at initiation but not originally required. This paper's proposal here tightens that specification: criteria for success becomes a required field for any PIFR whose maturity marker is set above the baseline *hypothesis* level. A hypothesis, honestly labelled, does not yet need to know how it will be tested. A claim marked *working theory*, *tested*, or *established* does — otherwise the marker is doing exactly what this section warns against, only for itself: confident-sounding structure with nothing behind it, one level further up the same failure mode.

In keeping with this principle, this paper itself should be marked accordingly: the central claim — that feedforward is structurally necessary for succession in a way feedback cannot be — is a hypothesis, strongly motivated by formal cybernetic theory and by an extended period of lived practice, but not yet tested under controlled conditions.

VIII. An Illustrative Case

A small exploratory experiment offers a concrete, if preliminary, illustration of the mechanism described above, rather than evidence for it in any statistically rigorous sense.

Three independent instances of a large language model were given the same creative task — producing a brochure for a community beach cleanup — under three different conditions. The first instance received only the task itself. The second received the task plus a continuity substrate describing originating intention, audience framing, and anti-drift constraints. The third received the task and a small set of verified conjectures derived from a reflective review of the second instance's reasoning — but none of the second instance's actual outputs, images, or analysis.

The first instance converged rapidly on a generic, symbolically polished representation — a recognisable default attractor, in the absence of any feedforward structure to orient it otherwise. The second instance, given a continuity substrate, shifted toward situated, locally realistic representation without losing creative variation — the substrate bounded interpretation without eliminating it. The third instance, despite receiving no direct output from the second, inherited the second instance's interpretive orientation: realism weighting, anti-symbolic framing, and situated stewardship — all while producing entirely independent content.

This is, in miniature, exactly what the Good Regulator Theorem predicts. The third instance was not given a copy. It was given a partial model — orientation, not output — and it functioned as a regulator capable of extending that model into new, unseen content. Nothing was accumulated forward. Something was succeeded forward.

The experiment is a single exploratory pilot in one domain, with no statistical power and no claim to generalisability. It is offered here as a worked illustration of the mechanism this paper proposes, not as proof of it.

IX. Limitations and the Honest Cost of Feedforward

Feedforward is not a strictly superior mechanism, and the cybernetics literature is clear about why. The disadvantage of feedforward regulation is that it is not always possible to act in time, and the anticipation may turn out to be incorrect, so that the action does not have the desired result.

A feedforward record that carries forward a flawed model of the original reasoning is, in a precise sense, more dangerous than no record at all — because it carries the authority of structure without the guarantee of accuracy. This is why the epistemic maturity marker proposed in Section VII is not an optional refinement. It is the mechanism by which a feedforward system protects itself against confidently transmitting an incorrect anticipation.

This is also why the most robust continuity architecture, consistent with the cybernetics literature more broadly, combines feedforward with feedback rather than replacing one with the other. PIFR-style feedforward artifacts reduce the need for reconstruction. PSS-style feedback architecture — authority membranes, mutation containment, inspectable accumulation — catches and corrects the cases where feedforward's anticipation was wrong. Neither is sufficient alone. The claim of this paper is narrower and more specific: that feedback alone, however well designed, cannot achieve succession, because succession requires a model transmitted in advance of the need for it, and feedback by definition only ever arrives after.

X. Research Agenda

This paper proposes a hypothesis grounded in established cybernetic theory and illustrated by a single exploratory case. What it does not yet have is controlled empirical validation, and naming the path to that validation honestly is part of the paper's obligation.

Future work would need to examine: comparative reconstruction cost in matched systems with and without feedforward artifacts; the relationship between the variety encoded in a feedforward record and the range of future situations it successfully anticipates; the effect of explicit epistemic maturity marking on how accurately future reasoning vehicles calibrate their confidence in inherited material; and replication of the cross-instance trajectory transfer observed in the illustrative case, across multiple domains, multiple model architectures, and larger sample sizes.

None of this exists yet. The hypothesis is offered in its current form because the cybernetic grounding is solid, the organisational parallel is independently documented, and the lived pattern — observed across many uncontrolled but high-stakes attempts — was consistent enough to be worth stating formally and testing properly.

XI. Conclusion

Continuity infrastructure built purely on feedback principles — however sophisticated, however well it lowers reconstruction cost — cannot achieve what this series has called continuity in the strongest sense, because feedback can only ever correct after a gap has opened. It cannot prevent the gap.

Feedforward can. Not by being more efficient at the same task, but by performing a structurally different one: transmitting a model of reasoning, not merely its conclusions, so that a future reasoning vehicle can anticipate rather than reconstruct.

The Good Regulator Theorem explains why this works, in formal terms that have stood since 1970. The organisational succession literature shows the same mechanism operating, independently discovered, in human institutions facing generational knowledge loss. A small exploratory experiment offers an early, illustrative glimpse of the same mechanism operating across independent AI reasoning instances.

The unit of continuity is not how much has been stored. It is whether what was understood can be succeeded — carried forward as a working model, oriented and ready to continue, rather than left behind as a conclusion to be painstakingly reconstructed.

We have done a great deal, across institutions, technologies, and decades, optimising for *more*.

This paper proposes that the way forward is *more sensible*.

Declarations

Author Affiliation and Research Programme

Independent researcher. Formal affiliation: ARTMA ApS. This paper is part of the Audited State Research Programme (auditedstate.app), which investigates Navigational Coherence — how understanding stays sufficiently coherent to support orientation, judgment, revision, and action as evidence, interpretations, and circumstances change — through a network of independently assessable papers rather than a linear series. This paper builds directly on Interpretive Entropy (Mayoh, 2026a), The Continuity Threshold (Mayoh, 2026b), Invisible Opportunity Cost (forthcoming), and Persistent Semantic Scaffolds and the Pay-It-Forward Record (forthcoming).

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

This paper was developed with the assistance of AI systems as a research and editorial tool, for drafting support, literature cross-referencing, and structural revision. The theoretical framework, arguments, and conclusions are the author's own.

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